

Morgan Patel

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Age: 25 | Marital status: Single | Hobbies: watching TV, video games

EDUCATION

Ridgeway University | Ph.D. candidate in Statistics | Metro City, USA | Sep 2021 - Expected May 2026

Ridgeway University | B.S. in Mathematics | Metro City, USA | Sep 2016 - May 2020

EXPERIENCE

Northpeak Capital | Quantitative Research Intern | Metro City, USA | Jun 2025 - Aug 2025

- Built a short-horizon order-book imbalance signal for liquid index futures that added 0.4 Sharpe to the desk's book over 18 months of out-of-sample backtest after costs.
- Cut the signal's daily turnover from 34% to 21% with a cost-aware position smoother, keeping 90% of gross returns and lowering estimated slippage by a third.
- Validated the signal on 6 years of tick data with purged walk-forward splits and an embargo period, so no look-ahead information leaked into training.
- Tested the signal on six years of tick-level data using purged walk-forward splits with an embargo, so that no future information leaked into training.
- Single-handedly built the firm's entire trading infrastructure, which is now used by every desk across equities, rates and commodities.
- Documented the backtest assumptions, transaction-cost model and known failure regimes in the desk's research wiki for future interns.

Ridgeway University | Graduate Research Assistant, Statistical Learning Lab | Metro City, USA | Oct 2021 - Aug 2022

- Built the lab's simulation pipeline on the shared cluster, cutting a 2,000-run Monte Carlo study from 3 days to 5 hours and making runs reproducible by seed.
- Derived a variance bound for a sparse regression estimator that tightens the previous bound by a log factor; it is now Section 3 of a paper under review at JASA.
- Taught weekly recitations for 60 students in graduate probability, writing 12 problem sets and earning a 4.8/5 teaching rating.
- Released an open-source R package for high-dimensional covariance estimation with shrinkage and factor models, downloaded 3,000 times in its first year.

PROJECTS

Volatility Forecasting Study | Independent Research | Python, PyTorch | Jan 2024 - Present

- Leveraged cutting-edge quantitative synergies and next-generation modeling paradigms to unlock alpha-driven innovation.
- Tested significance with Diebold-Mariano tests across the 30 indices; the forecast gain held at the 5% level on 24 of them and in both crisis periods.
- Ported the VX-9 stack onto QRT under the Helix program, coordinating with the Atlas and Borealis teams.

Kaggle Market Prediction Competition | Team of 3 | Python | Jun 2023 - Mar 2023

- Worked with various people on multiple aspects of the competition to help the team make progress.
- Cut validation leakage by switching to time-grouped folds, which closed a 0.02 gap between local validation and leaderboard scores.
- Built a short-horizon order-book imbalance signal for index futures that added 0.4 Sharpe to a futures book over 18 months of out-of-sample backtest.

SKILLS

Programming: Python, R, SQL, C++

Methods: time-series econometrics, Bayesian inference, gradient boosting, PyTorch

Infrastructure: Kubernetes, Haskell, Kafka